AFFORDABLE RENEWABLE ENERGY TRADING

Milestone: Project Proposal

Group 12

Siddiq Mohiuddin Mohammed

Ketaki Salway

857-565-8408

857-3911-895

salway.k@northeastern.edu

mohammed.si@northeastern.edu

Percentage of Effort Contributed by Student1: 50%

Percentage of Effort Contributed by Student2: 50%

Signature of Student 1:

Signature of Student 2:

Submission Date: 01/19/2025

AFFORDABLE RENEWABLE ENERGY TRADING

Ketaki Salway and Siddiq Mohiuddin Mohammed

Access to affordable and reliable electricity remains a challenge in underserved regions. While solar panels provide renewable energy, uneven distribution leaves some with excess energy and others with shortages, wasting potential and limiting broader benefits. Our project introduces a decentralized, peer-to-peer (P2P) energy trading platform to address this issue. Using a transformative "energy box," users connect to a local grid to buy and sell surplus energy seamlessly, fostering equitable access and a thriving local energy economy. Robust database systems will manage user information, track transactions, and ensure platform integrity, enabling accurate and secure energy trading. By redistributing surplus energy and maximizing renewable resources, this solution empowers communities with sustainable, reliable electricity while promoting economic growth and the efficient use of clean energy.

Theory

Availability of affordable and reliable electricity is a persistent challenge in many underprivileged regions. Although solar panels offer a sustainable energy solution, their benefits are often unevenly distributed. Some households or businesses generate more electricity than they can use, leading to waste, while others experience shortages due to limited capacity, excess requirement or unfavorable conditions. This disparity hinders the potential of solar energy to transform communities and creates barriers to reliable energy access.

Central to this concept is an innovative "energy box," a device that connects all users to a shared local grid, allowing them to exchange electricity effortlessly. This grid allows users to trade electricity directly without relying on centralized energy suppliers.

A reliable database system serves as the foundation of this initiative, facilitating the efficient management of user data, energy transactions, and grid activity. It ensures secure and accurate data processing, which is crucial for developing a scalable and dependable trading platform. Core functionalities include storing and updating user data such as energy production, consumption, and connection status, tracking the generation, consumption, and surplus of electricity in real time.

Additionally, the project encourages greater adoption of renewable energy by making solar panel investments more economically attractive. This approach fosters a sustainable energy ecosystem where even small-scale solar producers can contribute meaningfully to their communities.

In summary, this P2P energy trading platform integrates renewable energy utilization with advanced database systems to address energy access challenges in underserved regions. By redistributing surplus energy and creating a local energy marketplace, the project paves the way for equitable electricity distribution, economic development, and increased adoption of clean energy solutions.

Other Requirements

1. A user can participate in zero to infinite energy transactions as a buyer or seller; an energy transaction must involve exactly one buyer and one seller.
2. A user can have exactly one energy profile; an energy profile must belong to exactly one user.
3. A user can connect to zero to infinite other users in the same local grid; a user in the local grid can also connect to zero to infinite other users.
4. A region contains multiple users (households and businesses), but a user must belong to exactly one region.
5. A region must have exactly one supervisor; a supervisor must manage exactly one region.
6. A supervisor can handle zero to infinite complaints; a complaint must be managed by exactly one supervisor.
7. A user can submit zero to infinite complaints; a complaint must be submitted by exactly one user.
8. An energy transaction corresponds to exactly one payment; a payment must be associated with exactly one energy transaction.